

7-8b

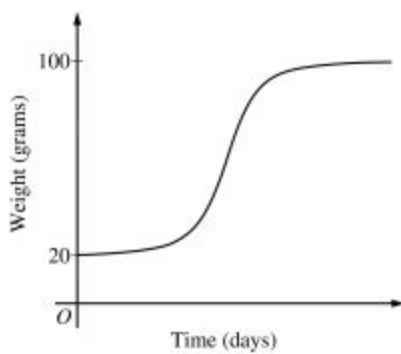
1. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

(a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

(b) Find $\frac{d^2B}{dt^2}$ in terms of B . Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



(c) Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

Part A

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.



0	1	2
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The student earns all of the following points:

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$$\left. \frac{dB}{dt} \right|_{B=40} = \frac{1}{5}(60) = 12$$

$$\left. \frac{dB}{dt} \right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left. \frac{dB}{dt} \right|_{B=40} > \left. \frac{dB}{dt} \right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

Part B

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5} \frac{dB}{dt} = -\frac{1}{5} \cdot \frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.



0	1	2
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Therefore, the graph of B is concave down for $20 \leq B < 100$. A portion of the given graph is concave up.

Part C

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1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$

$$-\ln|100 - B| = \frac{1}{5}t + c$$

Because $20 \leq B < 100$, $|100 - B| = 100 - B$.

$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$

$$100 - B = 80e^{-t/5}$$

$$B(t) = 100 - 80e^{-t/5}, t \geq 0$$



0	1	2	3	4	5
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$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

$$\int \frac{1}{100 - B} dB = \int \frac{1}{5} dt$$

$$-\ln|100 - B| = \frac{1}{5}t + c$$

Because $20 \leq B < 100$, $|100 - B| = 100 - B$.

$$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$$

$$100 - B = 80e^{-t/5}$$

$$B(t) = 100 - 80e^{-t/5}, t \geq 0$$

2. If $\frac{dy}{dx} = 4y$ and if $y=4$ when $x=0$, then $y=$

(A) $4e^{4x}$



(B) e^{4x}

(C) $3+e^{4x}$

(D) $4+e^{4x}$

(E) $2x^2 + 4$

3. Bacteria in a certain culture increase at a rate proportional to the number present. If the number of bacteria doubles in three hours, in how many hours will the number of bacteria triple?

(A) $3\ln 3/\ln 2$



(B) $2\ln 3/\ln 2$

(C) $\ln 3/\ln 2$

(D) $\ln(27/2)$

(E) $\ln(9/2)$

4.  Population y grows according to the equation $\frac{dy}{dt} = ky$, where k is a constant and t is measured in years. If the population doubles every 10 years, then the value of k is

(A) 0.069



(B) 0.200

(C) 0.301

(D) 3.322

(E) 5.000

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5.  A puppy weighs 2.0 pounds at birth and 3.5 pounds two months later. If the weight of the puppy during its first 6 months is increasing at a rate proportional to its weight, then how much will the puppy weigh when it is 3 months old?
- (A) 4.2 pounds
- (B) 4.6 pounds ✓
- (C) 4.8 pounds
- (D) 5.6 pounds
- (E) 6.5 pounds
6.  During a certain epidemic, the number of people that are infected at any time increases at a rate proportional to the number of people that are infected at that time. If 1,000 people are infected when the epidemic is first discovered, and 1,200 are infected 7 days later, how many people are infected 12 days after the epidemic is first discovered?
- (A) 343
- (B) 1,343
- (C) 1,367 ✓
- (D) 1,400
- (E) 2,057
7. The weight of a population of yeast is given by a differentiable function y , where $y(t)$ is measured in grams and t is measured in days. The weight of the yeast population increases according to the equation $\frac{dy}{dt} = ky$, where k is a constant. At time $t = 0$, the weight of the yeast population is 120 grams and is increasing at the rate of 24 grams per day.
- Which of the following is an expression for $y(t)$?
- (A) $120e^{24t}$
- (B) $120e^{t/5}$ ✓
- (C) $e^{t/5} + 119$
- (D) $24t + 120$

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8. Oil is being pumped continuously from a certain oil well at a rate proportional to the amount of oil left in the well; that is, $\frac{dy}{dt} = ky$, where y is the amount of oil left in the well at any time t . Initially there were 1,000,000 gallons of oil in the well, and 6 years later there were 500,000 gallons remaining. It will no longer be profitable to pump oil when there are fewer than 50,000 gallons remaining.

At what rate is the amount of oil in the well decreasing when there are 600,000 gallons of oil remaining?

Part B

1 point is earned for answer consistent with part a.

$$\begin{aligned}\frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2\end{aligned}$$

Decreasing at $10^5 \ln 2$ gal/year



0	1
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The student response earns one of the following points:

1 point is earned for answer consistent with part a.

$$\begin{aligned}\frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2\end{aligned}$$

Decreasing at $10^5 \ln 2$ gal/year

9.  The population P of a city grows according to the differential equation $\frac{dP}{dt} = kP$, where k is a constant and t is measured in years. If the population of the city doubles every 12 years, what is the value of k ?

(A) 0.058

(B) 0.061

(C) 0.167

(D) 0.693

(E) 8.318



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10. If $\frac{dy}{dt} = ky$ and k is a nonzero constant, then y could be
- (A) $2e^{kty}$
- (B) $2e^{kt}$ ✓
- (C) e^{kt+3}
- (D) $kty+5$
- (E) $\frac{1}{2}ky^2 + \frac{1}{2}$
11. An investment account is opened with 5000 dollars at time $t = 0$. The growth of the money in the account is modeled by the differential equation $\frac{dA}{dt} = 0.05A$, where $A(t)$ is the amount of money in the account in dollars and t is the number of years since the account was opened.
- Which of the following is an expression for $A(t)$?
- (A) $0.05t + 5000$
- (B) $0.05t^2 + 5000$
- (C) $e^{0.05t} + 4999$
- (D) $5000e^{0.05t}$ ✓

Answer D

Correct. The exponential growth and decay model $\frac{dy}{dt} = ky$, with initial condition $y = y_0$ when $t = 0$, has solutions of the form $y = y_0e^{kt}$. Here the dependent variable y is given by the function $A(t)$, $k = 0.05$, and $y_0 = A(0) = 5000$, so the solution is $A(t) = 5000e^{0.05t}$.

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12. During optimal conditions, the rate of change of the population of a certain organism is proportional to the population at time t , in hours. At time $t = 0$ hours, the population is 300. At time $t = 24$ hours, the population is 1000. At what time t is the population 500 ?

(A) $t = \frac{24\sqrt{2}}{\sqrt{7}}$

(B) $t = \frac{48}{7}$

(C) $t = \frac{24 \ln 500}{\ln 1000}$

(D) $t = \frac{\ln(\frac{5}{3})}{\frac{1}{24} \ln(\frac{10}{3})}$

**Answer D**

Correct. Let y be the population. The model for exponential growth and decay that arises from the statement “the rate of change of a quantity is proportional to the size of the quantity” is $\frac{dy}{dt} = ky$. The solution to this differential equation is $y = Ce^{kt}$. The information about the values of y at $t = 0$ and $t = 24$ is used to find the values of C and k .

$$\text{At } t = 0, y = 300 \Rightarrow C = 300.$$

$$\text{At } t = 24, y = 1000 \Rightarrow 1000 = 300e^{k(24)} \Rightarrow k = \frac{1}{24} \ln\left(\frac{1000}{300}\right) = \frac{1}{24} \ln\left(\frac{10}{3}\right).$$

The positive value of k is consistent with the assumption that the population is growing. The model $y = Ce^{kt}$ with $C = 300$ and $k = \frac{1}{24} \ln\left(\frac{10}{3}\right)$ is used to solve for t when $y = 500$.

$$500 = 300e^{kt} \Rightarrow t = \frac{\ln(\frac{5}{3})}{\frac{1}{24} \ln(\frac{10}{3})}$$

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13. During a chemical reaction, the rate of change of the amount of the chemical remaining is proportional to the amount remaining. At time $t = 0$, the amount of the chemical is 12 moles. At time $t = 4$, the amount of the chemical is 4 moles. At what time t is the amount of the chemical 3 moles? (A mole is a unit of measure used in chemistry.)

(A) $3\sqrt{2}$

(B) $\frac{9}{2}$

(C) $\frac{4 \ln 3}{\ln 4}$

(D) $\frac{4 \ln 4}{\ln 3}$

**Answer D**

Correct. Let y be the amount of the chemical. The model for exponential growth and decay that arises from the statement “the rate of change of a quantity is proportional to the size of the quantity” is $\frac{dy}{dt} = ky$. The solution to this differential equation is $y = Ce^{kt}$. The information about the values of y at $t = 0$ and $t = 4$ are used to find the values of C and k .

$$\text{At } t = 0, y = 12 \Rightarrow C = 12.$$

$$\text{At } t = 4, y = 4 \Rightarrow 4 = 12e^{4k} \Rightarrow k = \frac{1}{4} \ln \left(\frac{1}{3} \right) = -\frac{1}{4} \ln 3.$$

The negative value of k is consistent with the observation that the chemical is decaying. The model $y = Ce^{kt}$ with $C = 12$ and $k = -\frac{1}{4} \ln 3$ is used to solve for t when $y = 3$.

$$3 = 12e^{tk} \Rightarrow t = \frac{1}{k} \ln \left(\frac{1}{4} \right) = -\frac{1}{k} \ln 4 = \frac{4 \ln 4}{\ln 3}$$

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14.  A scientist monitored the number of fish in a lake and found that the number of fish increased at a rate proportional to the number of fish present. At time $t = 0$ years, there were 200 fish in the lake. At time $t = 2$ years, there were 1000 fish in the lake. Which of the following is the best estimate for the number of fish in the lake at time $t = 4$ years?
- (A) 1800
(B) 3400
(C) 5000 ✓
(D) 642,000

Answer C

Correct. Let N be the number of fish. The model for exponential growth that arises when the rate of change of a quantity is proportional to the size of the quantity is $\frac{dN}{dt} = kN$, where k is a constant. The solution to this differential equation is $N = Ce^{kt}$ for some constant C . The information about the values of N at $t = 0$ and $t = 2$ can be used to find the values of C and k .

At $t = 0$, $N = Ce^0 = 200 \Rightarrow C = 200$.

At $t = 2$, $1000 = 200e^{k(2)} \Rightarrow k = \frac{\ln 5}{2} \approx 0.805$.

Therefore, at $t = 4$, $N \approx 200e^{0.805(4)} \approx 5005 \approx 5000$.

15. The decreasing function M represents the mass of a sample of a radioactive element at time t . The rate of change of the sample's mass is proportional to the mass of the sample. Which of the following differential equations could be used to model the relationship between the mass of the sample and the rate of change of the sample's mass, where k is a constant?
- (A) $\frac{dM}{dt} = kt$
(B) $\frac{dM}{dt} = kM$ ✓
(C) $\frac{dM}{dt} = \frac{k}{t}$
(D) $\frac{dM}{dt} = \frac{k}{M}$

Answer B

Correct. The model for exponential growth and decay that arises when the rate of change of a quantity is proportional to the quantity is $\frac{dM}{dt} = kM$. Since the sample's mass is decreasing, this is an exponential decay model and the proportionality constant k would be negative.

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16. Let $P(t)$ represent the amount of a certain drug, in milligrams, that is in the bloodstream at time t hours. The rate at which the drug leaves the bloodstream is proportional to the amount of the drug in the bloodstream. The amount of the drug in the bloodstream can be modeled by $\frac{dP}{dt} = kP$, where k is a constant and t is time, in hours. At time $t = 0$, 500 milligrams of the drug was administered. At the moment when the amount of the drug in the bloodstream is 50 milligrams, the amount is decreasing at a rate of 11.5 milligrams per hour. Which of the following is an expression for $P(t)$?
- (A) $P(t) = 500 - 0.23t$
- (B) $P(t) = 500 - \frac{0.23}{2}t^2$
- (C) $P(t) = 500e^{-0.23t}$ ✓
- (D) $P(t) = e^{-0.23t} + 499$

Answer C

Correct. The exponential decay model $\frac{dP}{dt} = kP$ with initial condition $P(0) = 500$ has a solution of the form $P(t) = 500e^{kt}$. When $P(t) = 50$, $\frac{dP}{dt} = -11.5$, since the amount of drug in the bloodstream is decreasing. $\frac{dP}{dt} = kP \Rightarrow -11.5 = 50k \Rightarrow k = \frac{-11.5}{50} = -0.23$. Therefore, $P(t) = 500e^{-0.23t}$.

17.  The quantity R , in grams, of a certain radioactive substance decreases according to the exponential decay model $\frac{dR}{dt} = -0.05R$, where t is measured in seconds. During an experiment, a scientist determines that the rate of decay of a second substance with the quantity S , in grams, can be represented by a linear model $\frac{dS}{dt} = -4$, where t is measured in seconds. If at time $t = 0$, $R(0) = 100$ and $S(0) = 125$, at what time t , in seconds, will there be equal quantities of both substances?
- (A) $t = 6.318$
- (B) $t = 6.329$
- (C) $t = 23.548$ ✓
- (D) $t = 31.197$

Answer C

Correct. The particular solution to the differential equation $\frac{dR}{dt} = -0.05R$ with the initial condition $R(0) = 100$ is $R(t) = 100e^{-0.05t}$. The particular solution to the differential equation $\frac{dS}{dt} = -4$ with the initial condition $S(0) = 125$ is $S(t) = 125 - 4t$. The solution to $100e^{-0.05t} = 125 - 4t$ for $t > 0$ is found with the calculator to be $t = 23.548$.

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18. A dose of 400 milligrams of a drug is administered to a patient. The amount of the drug, in milligrams, in the person's bloodstream at time t , in hours, is given by $A(t)$. The rate at which the drug leaves the bloodstream can be modeled by the differential equation $\frac{dA}{dt} = kA$, where k is a constant. Which of the following could be an expression for $A(t)$?

(A) $A(t) = 400e^{-0.3t}$



(B) $A(t) = e^{-0.3t} + 399$

(C) $A(t) = -3t + 400$

(D) $A(t) = -1.5t^2 + 400$

Answer A

Correct. The exponential growth model, $\frac{dy}{dt} = ky$, with initial condition $y = y_0$ when $t = 0$, has a solution of the form $y = y_0e^{kt}$. This expression is of that form for $y = A(t)$, where $y_0 = 400$ and $k = -0.3$, since the amount of the drug in the bloodstream is decreasing.

19. The value of a car t years after it is purchased is given by the decreasing function V , where $V(t)$ is measured in dollars. The rate of change of the car's value in dollars per year is proportional to the car's value. Which of the following differential equations could be used to model the value of the car, where k is a constant?

(A) $\frac{dV}{dt} = kt$

(B) $\frac{dV}{dt} = kV$



(C) $\frac{dV}{dt} = \frac{k}{t}$

(D) $\frac{dV}{dt} = \frac{k}{V}$

Answer B

Correct. The model for exponential growth and decay that arises from the statement "The rate of change of a quantity is proportional to the quantity" is $\frac{dy}{dt} = ky$. Since the car's value is decreasing, this is an exponential decay model and the proportionality constant k would be negative.

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20. As a glacier melts, the volume V of the ice, measured in cubic kilometers, decreases at a rate modeled by the differential equation $\frac{dV}{dt} = kV$, where t is measured in years. The volume of the glacier is 400 km^3 at time $t = 0$. At the moment when the volume of the glacier is 300 km^3 , the volume is decreasing at the rate of 15 km^3 per year. What is the volume V in terms of time t ?
- (A) $V = 400 - 0.05t$
- (B) $V = 400 - 0.025t^2$
- (C) $V = 400e^{-0.05t}$ ✓
- (D) $V = e^{-0.05t} + 399$

Answer C

Correct. The exponential decay model $\frac{dV}{dt} = kV$ with initial condition $V = 400$ when $t = 0$ has a solution of the form $V = 400e^{kt}$. When $V = 300$, $\frac{dV}{dt} = -15$ since the volume of the ice is decreasing.

$$\frac{dV}{dt} = kV \Rightarrow -15 = 300k \Rightarrow k = -\frac{15}{300} = -0.05$$

21. The amount of bacteria in a petri dish increases at a rate proportional to the amount present. At time $t = 0$, the amount of bacteria in the dish is 10 grams. At time $t = 2$, the amount of bacteria in the dish is 30 grams. What is the amount of bacteria in the dish at time $t = 6$?
- (A) 70 grams
- (B) 190 grams
- (C) 270 grams ✓
- (D) 9270 grams

Answer C

Correct. Let y be the amount of bacteria. The model for exponential growth that arises from the statement “the rate of change of a quantity is proportional to the size of the quantity” is $\frac{dy}{dt} = ky$. The solution to this differential equation is $y = Ce^{kt}$. The information about the values of y at $t = 0$ and $t = 2$ can be used to find the values of C and k . At $t = 0$, $y = 10 \Rightarrow C = 10$. At $t = 2$, $y = 30 \Rightarrow 30 = 10e^{2k} \Rightarrow k = \frac{1}{2}\ln 3$. Therefore at $t = 6$, $y = 10e^{(\frac{1}{2}\ln 3) \cdot 6} = 10e^{3\ln 3} = 10e^{\ln 27} = 10 \cdot 27 = 270$.

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22.  Extreme heat applied to a colony of microorganisms causes the size P of the colony, measured in grams, to decrease according to the exponential decay model $\frac{dP}{dt} = -0.4P$, where the time t is measured in hours. The size Q of a second colony of microorganisms, also measured in grams, decreases at the constant rate of 1 gram per hour according to the linear model $\frac{dQ}{dt} = -1$. If at time $t = 0$ the first colony has size $P(0) = 2$ and the second colony has size $Q(0) = 3$, at what time will both colonies have the same size?
- (A) 1.437
(B) 1.667
(C) 2.156 ✓
(D) 2.654

Answer C

Correct. The particular solution to the differential equation $\frac{dP}{dt} = -0.4P$ with initial condition $P(0) = 2$ is $P(t) = 2e^{-0.4t}$. The particular solution to the differential equation $\frac{dQ}{dt} = -1$ with initial condition $Q(0) = 3$ is $Q(t) = 3 - t$. The solution to $2e^{-0.4t} = 3 - t$ for $t > 0$ is found with a calculator to be $t \approx 2.156$.

23. A kitten weighs 85 grams at birth. During the first four weeks after the kitten's birth, its weight in grams is given by the function W that satisfies the differential equation $\frac{dW}{dt} = kW$, where t is measured in days and k is some positive constant. Which of the following could be an expression for $W(t)$?
- (A) $85e^{0.059t}$ ✓
(B) $4e^{0.162t} + 81$
(C) $13t + 85$
(D) $0.466t^2 + 85$

Answer A

Correct. The exponential growth model, $\frac{dy}{dt} = ky$, with initial condition $y = y_0$ when $t = 0$, has a solution of the form $y = y_0e^{kt}$. This expression is of that form for the function $y = W(t)$, with $y_0 = 85$ and $k = 0.059$.

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24. Oil is being pumped continuously from a certain oil well at a rate proportional to the amount of oil left in the well; that is, $dy/dt = ky$, where y is the amount of oil left in the well at any time t . Initially there were 1,000,000 gallons of oil in the well, and 6 years later there were 500,000 gallons remaining. It will no longer be profitable to pump oil when there are fewer than 50,000 gallons remaining.
- (a) Write an equation for y , the amount of oil remaining in the well at any time t .
- (b) At what rate is the amount of oil in the well decreasing when there are 600,000 gallons of oil remaining?
- (c) In order not to lose money, at what time t should oil no longer be pumped from the well?

Part A

2 points are earned for $y = e^{kt}$ **or** 1 point for each: $\ln|y| = kt + C$, $y = e^{kt} + C$

1 point is earned for $C = 10^6$

1 point is earned for $k = -\frac{\ln 2}{6}$

1 point is earned for answer

*Max 2/5 If y is not an exponential

$$\frac{dy}{dt} = ky \quad \text{or} \quad \begin{cases} \frac{dy}{y} = k dt \\ \ln|y| = kt + C_1 \\ y = e^{kt+C_1} \end{cases}$$

$$t = 0 \Rightarrow C = 10^6, C_1 = \ln 10^6$$

$$\therefore y = 10^6 e^{kt}$$

$$t = 6 \Rightarrow \frac{1}{2} = e^{6k}$$

$$\therefore k = -\frac{\ln 2}{6}$$

$$y = 10^6 e^{\frac{-t}{6} \ln 2} = 10^6 \cdot 2^{\frac{-t}{6}}$$



0	1	2	3	4	5
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The student response earns all of the following points:

7-8b

2 points are earned for $y = Ce^{kt}$ **or** 1 point for each: $\ln|y| = kt + C$, $y = e^{kt} + C$

1 point is earned for $C = 10^6$

1 point is earned for $k = -\frac{\ln 2}{6}$

1 point is earned for answer

*Max 2/5 If y is not an exponential

$$\frac{dy}{dt} = ky \quad \text{or} \quad \begin{cases} \frac{dy}{y} = k dt \\ \ln|y| = kt + C_1 \\ y = e^{kt+C_1} \end{cases}$$

$$t = 0 \Rightarrow C = 10^6, C_1 = \ln 10^6$$

$$\therefore y = 10^6 e^{kt}$$

$$t = 6 \Rightarrow \frac{1}{2} = e^{6k}$$

$$\therefore k = -\frac{\ln 2}{6}$$

$$y = 10^6 e^{\frac{-t}{6} \ln 2} = 10^6 \cdot 2^{\frac{-t}{6}}$$

Part B

1 point is earned for answer consistent with part a.

$$\begin{aligned} \frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2 \end{aligned}$$

Decreasing at $10^5 \ln 2$ gal/year



0	1
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The student response earns all of the following points:

7-8b

1 point is earned for answer consistent with part a.

$$\begin{aligned}\frac{dy}{dt} &= ky = -\frac{\ln 2}{6} \cdot 6 \cdot 10^5 \\ &= -10^5 \ln 2\end{aligned}$$

Decreasing at $10^5 \ln 2$ gal/year

Part C

1 point is earned for solving $5 \cdot 10^4$

1 point is earned for $kt = -\ln 20$

1 point is earned for answer

$$5 \cdot 10^4 = 10^6 e^{kt}$$

$$\therefore kt = -\ln 20 \left[= \ln \frac{1}{20} \right]$$

$$\therefore t = \frac{-\ln 20}{\frac{-\ln 2}{6}}$$

$$= 6 \frac{\ln 20}{\ln 2} = 6 \log_2 20$$

$$6 \frac{\ln 20}{\ln 2} \text{ years after starting}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for solving $5 \cdot 10^4$

1 point is earned for $kt = -\ln 20$

1 point is earned for answer

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$$5 \cdot 10^4 = 10^6 e^{kt}$$

$$\therefore kt = -\ln 20 \left[= \ln \frac{1}{20} \right]$$

$$\therefore t = \frac{-\ln 20}{\frac{-\ln 2}{6}}$$

$$= 6 \frac{\ln 20}{\ln 2} = 6 \log_2 20$$

$$6 \frac{\ln 20}{\ln 2} \text{ years after starting}$$

25. Which of the following is the solution to the differential equation $\frac{dy}{dx} = -3y$ with the initial condition $y(0) = 7$?

(A) $y = 7e^{-3x}$



(B) $y = e^{-3x} + 6$

(C) $y = -\frac{3}{2}x^2 + 7$

(D) $y = \sqrt{49 - 6x}$

Answer A

Correct. The exponential growth and decay model, $\frac{dy}{dx} = ky$, with initial condition $y = y_0$ when $x = 0$, has solutions of the form $y = y_0 e^{kx}$. Here, $k = -3$ and $y_0 = 7$. Therefore, the solution is $y = 7e^{-3x}$.

The differential equation can also be solved by using separation of variables and the initial condition to determine the appropriate value for the arbitrary constant.

$$\frac{dy}{dx} = -3y \Rightarrow \frac{1}{y} dy = -3 dx$$

$$\int \frac{1}{y} dy = \int -3 dx \Rightarrow \ln y = -3x + C \Rightarrow y = e^{-3x+C} = e^C e^{-3x} = k e^{-3x}$$

$$7 = k e^0 = k$$

$$y = 7e^{-3x}$$

Note: Absolute value was not used in the antiderivative of $\frac{1}{y}$, since the initial value of y is positive.

7-8b

26. $\frac{d}{dx}(x^3 \sec(2x)) =$

(A) $6x^2 \sec(2x) \tan(2x)$

(B) $2x^3 \tan^2(2x) + 3x^2 \sec(2x)$

(C) $x^3 \sec(2x) \tan(2x) + 3x^2 \sec(2x)$

(D) $2x^3 \sec(2x) \tan(2x) + 3x^2 \sec(2x)$

**Answer D**

Correct. A combination of the product rule and the chain rule is used to compute the derivative.

$$\frac{d}{dx}(x^3 \sec(2x)) = x^3 \cdot \frac{d}{dx}(\sec(2x)) + 3x^2 \cdot \sec(2x)$$

$$= x^3 \cdot (\sec(2x) \tan(2x) \cdot 2) + 3x^2 \sec(2x)$$

$$= 2x^3 \sec(2x) \tan(2x) + 3x^2 \sec(2x)$$